NHL Goal Prediction

Does spatial deep learning outperform classical expected goals models?

The Problem

In hockey analytics, not all shots are created equal.

A shot from the slot directly in front of the net is far more dangerous than a low-percentage attempt from the blue line. Quantifying that difference is the foundation of expected goals (xG) modeling.

The core question:
Can a CNN that explicitly models spatial structure on the ice outperform classical approaches that rely on engineered features like distance and angle?

Github Link

Does where you shoot matter more than how you shoot?

Three Models, One Question

Each model adds capacity. The question is whether that capacity translates to better predictions.

01

Logistic Regression

Classic xG baseline

Distance + angle only. Linear relationship assumed. Surprisingly strong.

02

MLP

Tabular nonlinear model

Same features, more capacity. Can it learn distance × angle interactions?

03

Spatial CNN

Grid-based rink representation

Treat the ice as an image. Let the network learn scoring geometry directly.

The Data

528,720

shot attempts
2020–2025 seasons

7%

goal rate
significant class imbalance

2

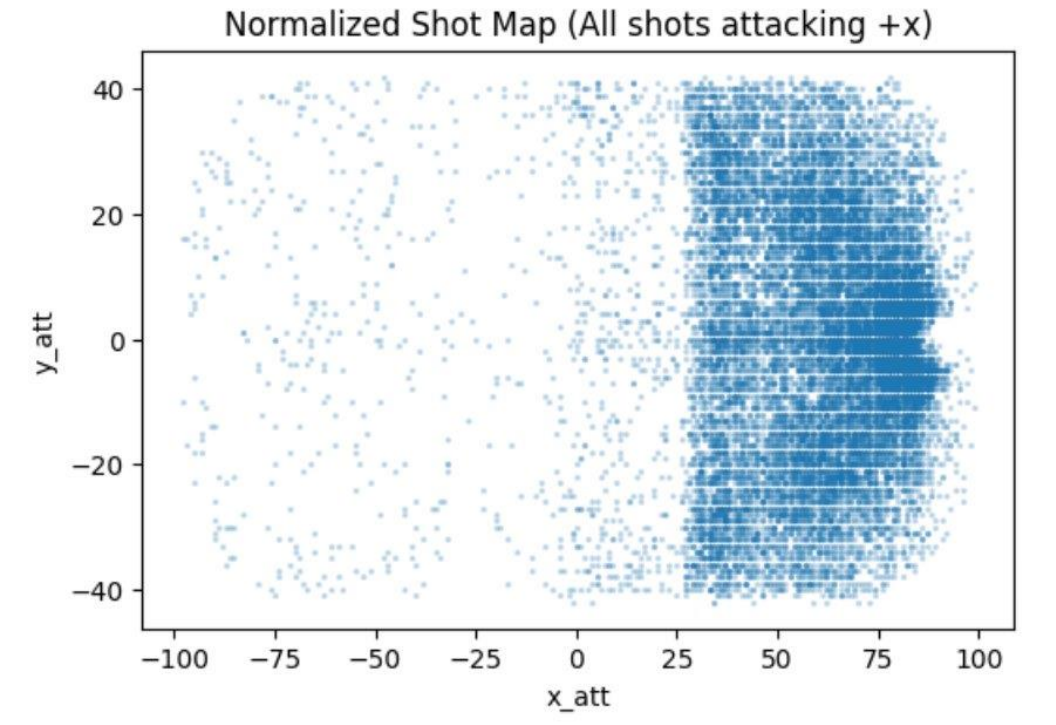
core features
distance + angle

Key preprocessing decisions

Shot direction normalized	All attacks toward the same goal; essential for CNN spatial consistency
Class-weighted loss	Compensates for 7% goal rate without oversampling
Time-based split	Train: 2020–2024 | Test: 2024–2025 season
StandardScaler	Applied to LR and MLP features | CNN uses raw grid coordinates

EDA

Normalized shot map (all attacks → +x)



528,720

total shot attempts

7%

goal rate · strong class imbalance

The dense concentration in the offensive zone confirms that shot location is highly structured — not random across the ice. The normalization step (all attacks toward +x) is what makes CNN spatial learning possible.

Logistic Regression The Baseline

0.709

ROC-AUC

0.143

PR-AUC

Features: distance + angle only

Assumes linear log-odds relationship
No feature interactions
Treats rink as feature space, not geometry

Interpretation

ROC-AUC of 0.709 is surprisingly strong for just two features. Distance and angle already capture most of the signal in goal probability. Goals are more likely closer to the net and from central positions, and that relationship is mostly monotonic.

This makes logistic regression a hard baseline to beat. The nonlinear models need to find signals that two summary statistics missed.

MLP Nonlinear Tabular Model

0.710
ROC-AUC

0.152
PR-AUC

vs. Logistic Regression

ROC-AUC +0.001 (marginal)
PR-AUC +0.010 (meaningful)
Training cost: low

Interpretation

The MLP adds nonlinear capacity over logistic regression, allowing it to model interactions between distance and angle. The modest ROC-AUC gain (+0.001) suggests the linear assumption wasn't far off for ranking shots.

The PR-AUC gain (+0.010) is more meaningful; under class imbalance, better precision on the positive class matters. But the feature space is still limited to two engineered variables. The MLP doesn't explicitly use spatial structure.

Spatial CNN: 3-Channel CNN Success

1-Channel CNN — ROC-AUC: 0.658

Single spatial grid encoding shot frequency. Underperformed logistic regression.

Why this failed: one channel can't separate where shots happen from where goals happen. The model learned density, not danger.

3-Channel CNN — ROC-AUC: 0.715

Channel 1: Shot location grid
Channel 2: Distance map
Channel 3: Angle map

Each layer adds a different spatial signal. Now the network can learn danger, not just density.

The lesson: model performance is not just about depth or complexity.

The architecture was the same. The representation changed. That was enough to flip the CNN from worst performer to best. Input encoding matters as much as model capacity.

Hyperparameter Tuning

Learning Rate

1e-3 → 5e-4

ROC-AUC: 0.715 → 0.717

✓ **Meaningful gain**

More stable gradient updates. Biggest single improvement.

Model Depth

Baseline → Deeper arch

ROC-AUC: 0.715 → 0.716

— **Marginal**

Shot danger is spatially smooth. More layers don't help a simple geometry problem.

Dropout

0.4 → 0.3

ROC-AUC improved, PR-AUC didn't

— **Mixed**

Less regularization helped ROC-AUC but hurt precision on the positive class.

Final model: lr=5e-4, dropout=0.4, baseline depth — best PR-AUC, stable training.

Model Comparison

Model	ROC-AUC	PR-AUC	Training Cost
Logistic Regression	0.709	0.143	Very Low
MLP (Tabular)	0.710	0.152	Low
CNN (3-channel)	**0.718**	**0.152**	**Moderate**

Key finding

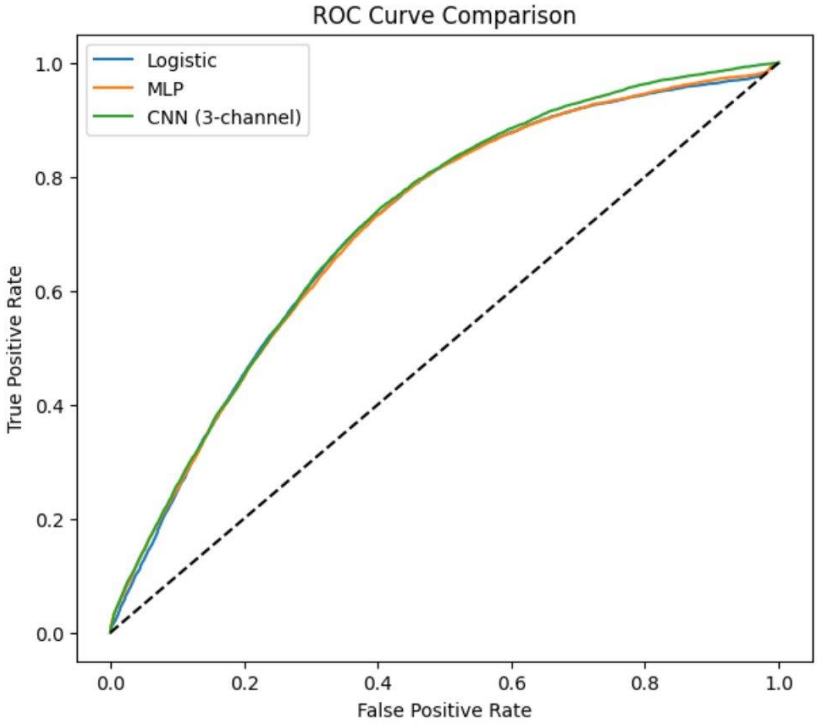
PR-AUC is the right metric. With 7% goal rate, precision on the positive class matters more than overall ranking.

Spatial CNN

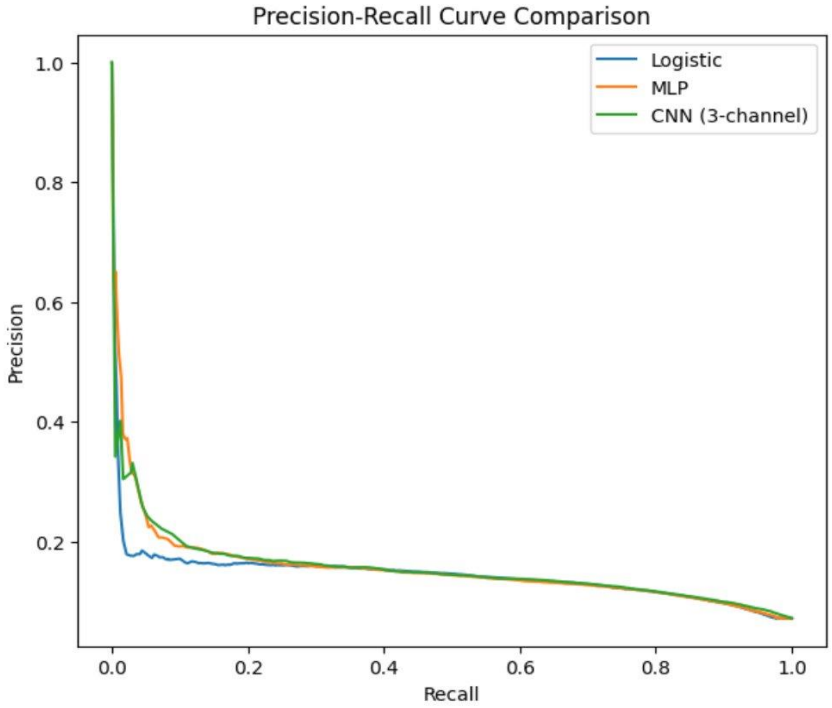
Best PR-AUC, but only once representation was fixed. The 1-channel version was the worst model.

Model Comparison: Results

ROC Curve



Precision-Recall Curve (more informative under 7% imbalance)



ROC curves are nearly overlapping. All three models rank shots similarly. The PR curve is where the CNN earns its edge: higher precision at low recall means fewer false alarms on high-confidence goal predictions.

Results

Final model scores

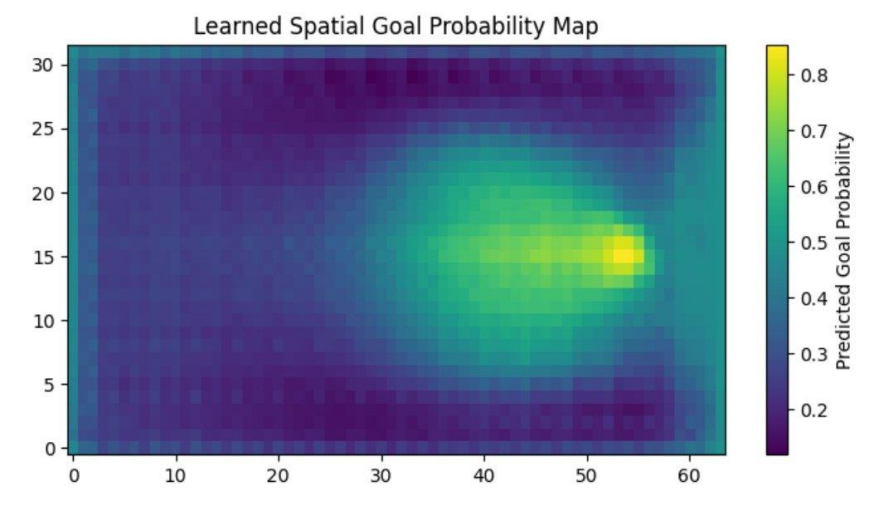
	Model	ROC-AUC	PR-AUC
0	Logistic Regression	0.709175	0.142731
1	MLP (Tabular)	0.709613	0.152441
2	CNN (3-channel)	0.717701	0.150594

+1.3%
ROC-AUC gain
LR → CNN

+5.5%
PR-AUC gain
LR → MLP

0.658
1-ch CNN
(worst model)

Learned spatial goal probability map



The CNN independently learned that the slot region (center-front) is the highest danger zone — without being told where the slot is. This validates the spatial encoding and confirms the model learned real hockey geometry.

Conclusions

Representation matters more than architecture.

The 1-channel CNN failed. The 3-channel CNN succeeded. Same network, different encoding. That's the most important finding.

Deep learning doesn't automatically win.

Logistic regression with two features hit ROC-AUC 0.709. The CNN needed significant work to surpass it, and the margin is modest.

Spatial structure is real and learnable.

The CNN's learned probability heatmap matches known hockey principles: slot shots are most dangerous, perimeter shots least. It found that geometry without being told.